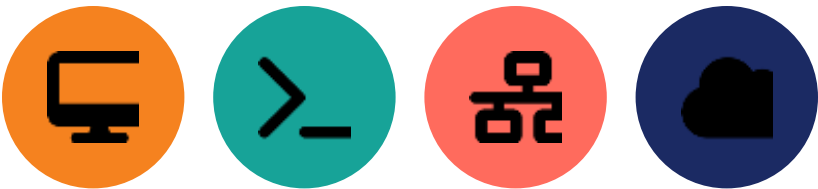


ORIENTATION TO COMPUTING • CSE111

UNIT 1

Computing Environment

From operating systems and Linux to networks, virtualization, and the cloud that powers it all



What is an Operating System?



What is an Operating System?



Software that manages all of a computer's hardware and software resources, and acts as the bridge between YOU and the machine.

Think of it as an air-traffic controller

The CPU, memory, storage, keyboard, and every app are like planes wanting the runway at once. The OS decides who goes when, so nothing collides and everything runs smoothly.

Without an OS...

Every app would need to speak directly to the hardware itself — no shared screen, no multitasking, no file system.

The Key Functions of an Operating System

Everything an OS does radiates from one job: managing resources



The OS Functions, Explained



Process Management

Decides which program gets the CPU, and for how long



Memory Management

Allocates and frees RAM so processes don't collide



File Management

Organises data into files & folders, controls access



Device Management

Talks to printers, disks, keyboards via drivers



Security

Enforces logins, permissions, and user isolation



User Interface

Gives you a GUI or command line to interact with it all

Types of Operating Systems

Batch OS

Jobs collected and run in groups, with no direct interaction

Examples include IBM's z/OS, IBM OS/360, and Unisys MCP

Early bank cheque processing systems

Time-Sharing OS

The CPU rapidly switches between multiple users' tasks

Examples Include UNIX, Multics, and Linux

Multi-user Linux/UNIX servers

Distributed OS

Multiple computers appear as a single system to users

Examples include Plan 9 from Bell Labs, Google Fuchsia OS, and Amoeba

Google's data-centre systems

Real-Time OS (RTOS)

Guarantees a response within a strict time limit

Examples include FreeRTOS, VxWorks, and QNX

Car airbags, pacemakers, industrial robots

Mobile OS

Built for touch, battery-life, and app ecosystems

Examples include Android, iOS, HarmonyOS, KaiOS

Mobile phones, Handhelds, Tablets.

Operating Systems Around You



Windows

Personal & office computers
worldwide



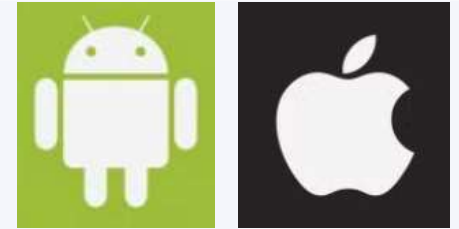
Linux

Servers, cloud infrastructure,
supercomputers



macOS

Apple desktops & laptops, creative
work



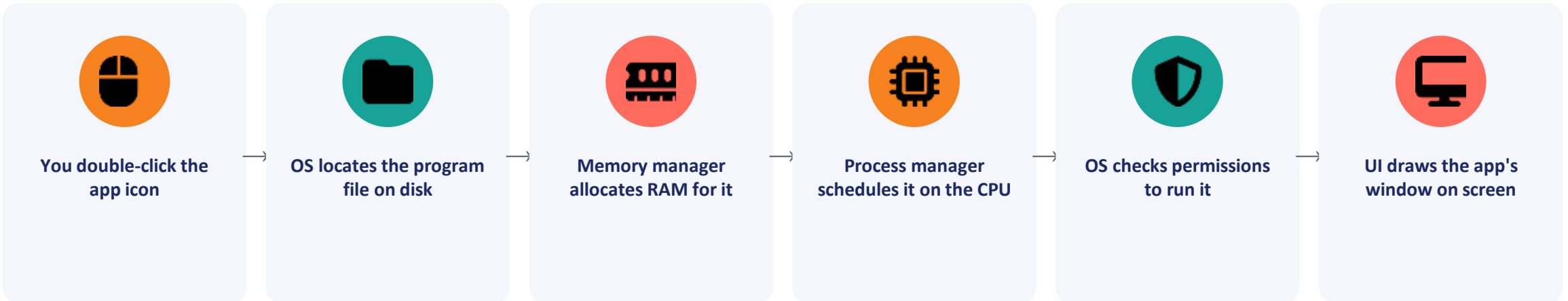
Android / iOS

Smartphones and tablets

Fun fact: most of the internet — including the servers hosting this course — runs on Linux.

Example: What Happens When You Open an App

A round trip through every OS function, in under a second



All of this — disk access, memory allocation, CPU scheduling, permission checks, and rendering — happens before you even see the app appear.



Try It Yourself: Which OS Type?



Match each scenario to the OS type it needs: a pacemaker, a university's shared server, and your smartphone.

Write the correct OS type next to each scenario:

Pacemaker →

Shared server →

Smartphone →

Try It Yourself: Which OS Type?



Match each scenario to the OS type it needs: a pacemaker, a university's shared server, and your smartphone.

Write the correct OS type next to each scenario:

Pacemaker → Real-time Operating System (RTOS)

Shared server → Multi-user Operating System (Time Sharing) / Distributed Operating System

Smartphone → Mobile Operating System

QUICK RECAP

Operating Systems

- The OS manages hardware and software so nothing collides
- Six core jobs: process, memory, file, device, security, and UI management
- Types range from batch and real-time to mobile and distributed
- Windows, Linux, macOS, and Android/iOS are OS families you use daily

Why Linux Matters for a CS Student



Runs the Internet

Most web servers, cloud platforms, and supercomputers run Linux



The Developer's Tool

Command-line skills transfer directly to real engineering jobs



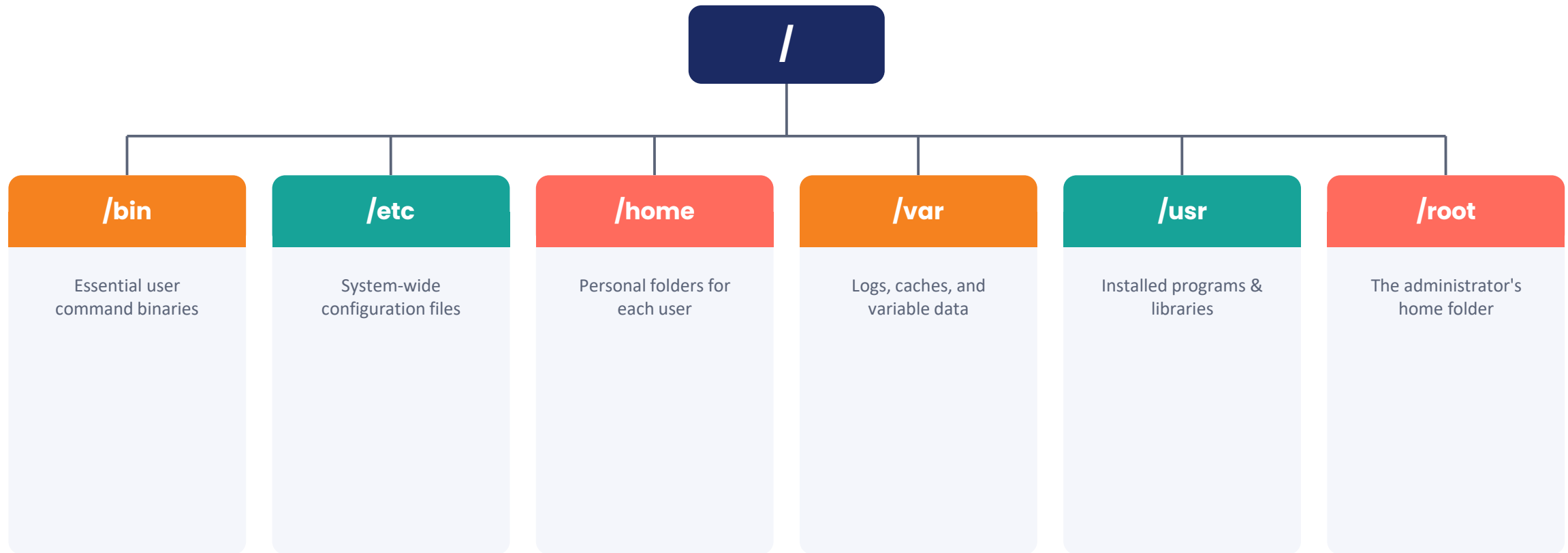
Free & Open Source

You can inspect, modify, and learn from the actual OS code

Over 96% of the world's top 1 million web servers run Linux, and it powers nearly all public cloud infrastructure.

The Linux File System Hierarchy

Everything in Linux lives in one tree, starting at the root: /



Navigation Commands

The five commands you'll type most often

pwd**Print Working Directory**

Shows the folder you're currently in

ls**List**

Shows files & folders in the current directory

cd**Change Directory**

Moves you into another folder

mkdir**Make Directory**

Creates a new folder

rmdir**Remove Directory**

Deletes an empty folder

File Management Commands

Creating, copying, moving, and viewing files



touch

Create an empty file



cp

Copy a file or folder



mv

Move or rename a
file



rm

Delete a file



cat

Display a file's
contents



chmod

Change file
permissions

Example: A Few Minutes in the Terminal

A realistic sequence of commands to set up a small project



```
$ pwd
```

```
/home/student
```

```
$ mkdir project
```

```
$ cd project
```

```
$ touch app.py notes.txt
```

```
$ ls
```

```
app.py  notes.txt
```

```
$ chmod +x app.py
```

```
$ cat notes.txt
```

```
This is my first Linux project!
```

Try It Yourself: Which Command?



You need to: (1) see your current folder, (2) create a folder called 'labs', (3) copy report.txt into it, (4) view its contents.

Write the correct Linux command for each step:

1.

2.

3.

4.

Try It Yourself: Which Command?



You need to: (1) see your current folder, (2) create a folder called 'labs', (3) copy report.txt into it, (4) view its contents.

Write the correct Linux command for each step:

1. `pwd` (View the current folder)
2. `mkdir labs` (Create a folder named **labs**)
3. `cp report.txt labs/` (Copy **report.txt** into the **labs** folder)
4. `ls labs` (View the contents of the **labs** folder)

QUICK RECAP

Linux File System & Commands

- Linux organises everything in one tree starting at the root /
- Key folders: /bin, /etc, /home, /var, /usr, /root
- Navigate with pwd, ls, cd, mkdir, rmdir
- Manage files with touch, cp, mv, rm, cat, chmod

Understanding WSL

- **Seamless Integration:** The Windows Subsystem for Linux (WSL) allows developers to run a GNU/Linux environment natively on Windows without dual-booting.
- **Developer Velocity:** Access Linux command-line tools, utilities, and applications alongside native Windows workflows.
- **Resource Efficiency:** Consumes far less memory and overhead compared to running a traditional, heavy Virtual Machine.
- **Architecture Evolution:** Transitioned from a translation layer (WSL 1) to a true, lightweight Hyper-V utility VM (WSL 2).



Installing **WSL** in Windows



1. Launch Terminal

Search for Windows PowerShell or Windows Terminal in your Start menu, right-click, and select "**Run as administrator**" to gain necessary privileges.



2. Execute Command

Type `wsl --install` and hit Enter.

This enables required virtual machine features and automatically downloads the default Ubuntu distribution.



3. Reboot & Config

Restart your machine. Upon reboot, a Linux console will open automatically to finalize the installation.

You will then set up your UNIX username and password.

What is a Computer Network?



A group of two or more computers linked together to share data, files, and resources like printers and internet access.



Share Resources

One printer or internet line, used by everyone on the network



Share Data

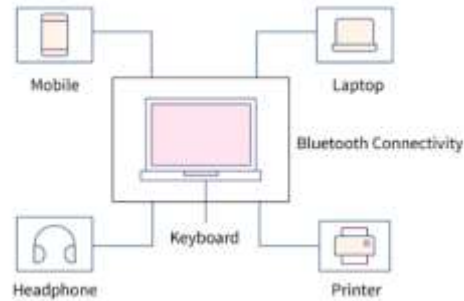
Files and databases accessible from any connected device



Communicate

Email, messaging, and video calls all travel over networks

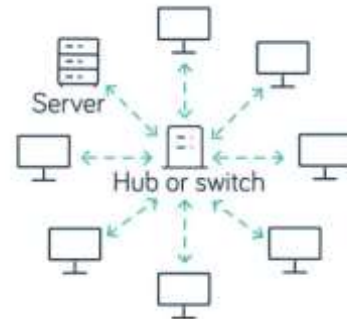
Types of Networks, by Size



Personal Area Network

A few metres — you and your devices

Bluetooth earbuds to your phone



Local Area Network

One building or campus

Your college computer lab



Metropolitan Area Network

A city

A city's cable TV / traffic-camera network



Wide Area Network

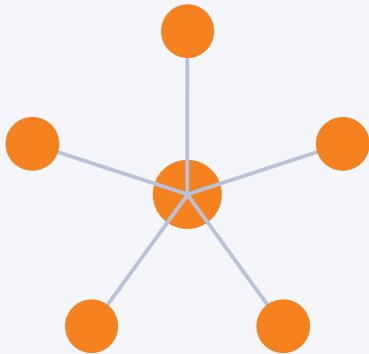
Countries or continents

The Internet itself

Network Topologies

The physical or logical layout of how devices connect

STAR



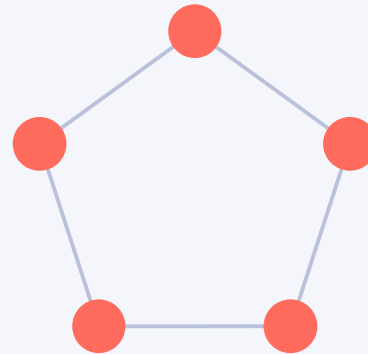
All devices connect to one central hub or switch

BUS



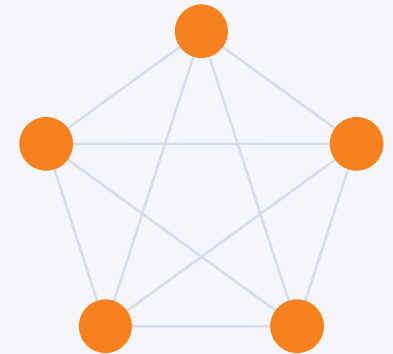
All devices share a single central cable

RING



Each device connects to exactly two neighbours

MESH



Devices connect to many or all others directly

Example: How Data Travels to a Website

What happens between typing a URL and seeing the page



All of this typically happens in well under a second — crossing PAN, LAN, and WAN boundaries along the way.

Client-Server Architecture



1. The Client Request

The client (e.g., a web browser or mobile app) initiates a connection and sends a formulated request over the network for specific data or a service.



2. Server Processing

A centralized server receives the request, validates it, queries the database if necessary, and executes the business logic required to formulate a response.



3. The Response

The server transmits the processed data (HTML, JSON, XML) back to the client, fulfilling the original request efficiently and securely.

Try It Yourself: Name That Network



Classify each as PAN, LAN, MAN, or WAN: a smartwatch paired to a phone, a hostel's Wi-Fi, and a video call with someone abroad.

Write the correct network type next to each scenario:

Smartwatch →

Hostel Wi-Fi →

Video call abroad →

Try It Yourself: Name That Network



Classify each as PAN, LAN, MAN, or WAN: a smartwatch paired to a phone, a hostel's Wi-Fi, and a video call with someone abroad.

Write the correct network type next to each scenario:

Smartwatch →

Personal Area Network (PAN)

Hostel Wi-Fi →

Local Area Network (LAN)

Video call abroad →

Wide Area Network (WAN)

QUICK RECAP

Computer Networks

- A network connects devices to share resources, data, and communication
- Sized from smallest to largest: PAN → LAN → MAN → WAN
- Topology is the layout: star, bus, ring, or mesh
- Real traffic crosses multiple network types on its way to you

Key Network Devices



Router

Directs traffic between different networks



Switch

Connects devices within one network intelligently



Hub

Connects devices by broadcasting to all of them



Modem

Converts signals between your ISP and your network



Access Point

Lets wireless devices join a wired network



NIC

The hardware inside a device that connects it to a network

Router vs. Switch vs. Hub

The classic point of confusion, cleared up



Router

Connects: Between networks (e.g. your home ↔ the Internet)

Intelligence: Smart — finds the best path



Switch

Connects: Within one network (e.g. office computers)

Intelligence: Smart — sends data only where needed



Hub

Connects: Within one small, old-style network

Intelligence: Dumb — broadcasts to every device

Example: Devices in a Home Network

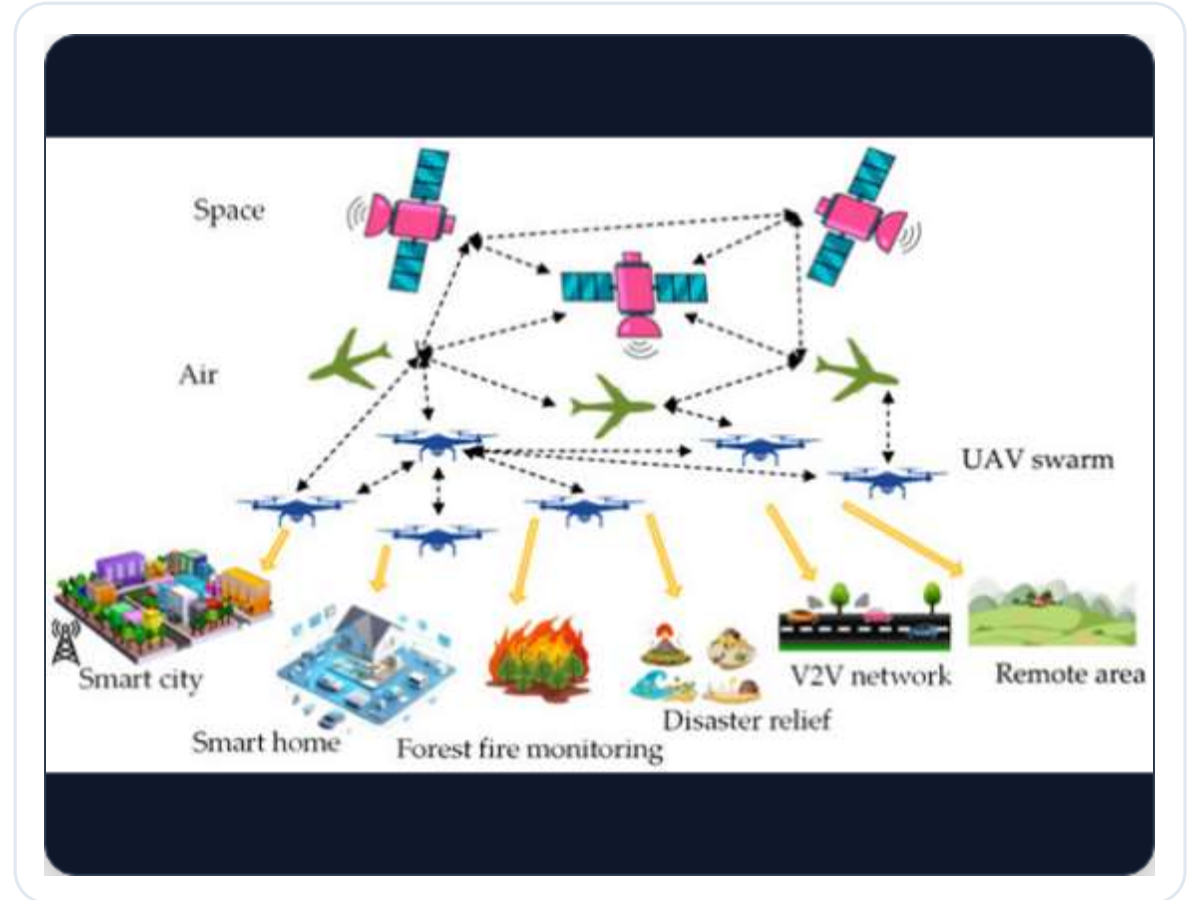
How your Wi-Fi actually gets you online



The modem connects your home to your ISP; the router shares that single connection wirelessly to every device you own.

Understanding Network Protocols

- **The Language of Networks:** A protocol is a standardized set of rules and formats that govern how data is transmitted, received, and understood across a network.
- **Ensuring Compatibility:** They ensure that devices from different manufacturers, running different operating systems, can communicate seamlessly.
- **Establishing Rules:** Protocols dictate error checking, data compression mechanisms, and the syntax of the communication.
- **Examples:** Common examples include HTTP for web traffic, SMTP for email, and TCP/IP as the foundational suite for the Internet.



The 7-Layer OSI Model

7



Application: Network process to application interfacing.

6



Presentation: Data formatting, compression, and encryption.

5



Session: Establishes, maintains, and terminates connections.

4



Transport: End-to-end data delivery and error recovery.

3



Network: Logical addressing (IP) and routing path determination.

2



Data Link: Physical addressing (MAC) and node-to-node transfer.

1



Physical: Binary transmission over copper, fiber, or wireless media.

Try It Yourself — Then Recap

Your Wi-Fi is slow because 20 devices are all fighting over one old hub instead of a switch. Why would replacing it with a switch help?

Your answer:

QUICK RECAP

- Routers connect networks, switches connect devices smartly, hubs just broadcast
- Modems bridge your home to your ISP; access points add Wi-Fi
- A typical home chain: Internet → Modem → Router → Your devices

Try It Yourself — Then Recap

Your Wi-Fi is slow because 20 devices are all fighting over one old hub instead of a switch. Why would replacing it with a switch help?

Answer: Replacing the **hub** with a **switch** improves network performance because a switch sends data **only to the intended device**, while a hub broadcasts data to **all devices**. This reduces network traffic and data collisions, allowing devices to communicate faster and making the Wi-Fi/network more efficient.

QUICK RECAP

- Routers connect networks, switches connect devices smartly, hubs just broadcast
- Modems bridge your home to your ISP; access points add Wi-Fi
- A typical home chain: Internet → Modem → Router → Your devices

What is a Hypervisor?



Software that lets one physical computer run several separate, independent “virtual” computers at the same time.



One Machine, Many OSes

Run Windows, Linux, and more, all on the same hardware



Safe Isolation

If one virtual machine crashes, the others keep running



Better Utilisation

Uses a physical server's full capacity instead of wasting it

Real World: Why Companies Use Hypervisors



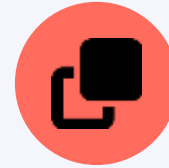
Server Consolidation

10 physical servers become 1 powerful server running 10 VMs — huge cost & energy savings



Safe Testing

Try risky updates or new software on a VM without touching the real system



Fast Disaster Recovery

A crashed VM can be restored from a snapshot in minutes, not days

Virtually every cloud data centre — AWS, Azure, Google Cloud — runs on hypervisors at massive scale.

What is a Virtual Machine?



“A computer inside a computer” — a software-based machine with its own virtual CPU, memory, disk, and OS, built by a hypervisor.

Physical Machine (Host)

Hypervisor

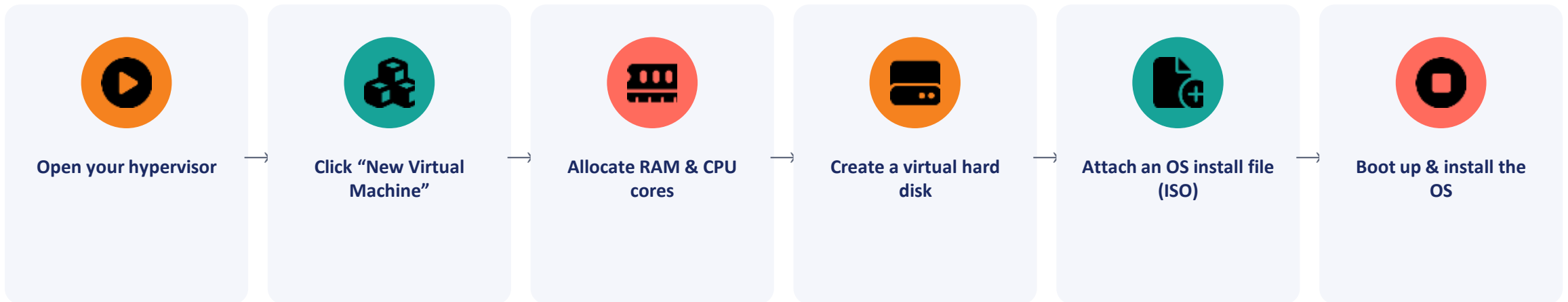
Guest OS: Windows

Guest OS: Linux

Guest OS: macOS

Creating a Virtual Machine

The basic steps, whether on VirtualBox, VMware, or a cloud console



Once created, the VM behaves exactly like a real computer — you can install software, browse the web, and even delete it without any risk to your real machine.

Managing a Virtual Machine



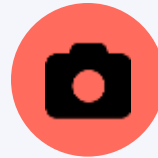
Start

Boot the VM up from where it left off



Stop

Shut the VM down cleanly



Snapshot

Save the VM's exact state to return to later



Clone

Make an identical copy of the whole VM



Delete

Permanently remove the VM and free its resources

Real World: Students & Developers Using VMs



Learning Linux Safely

Practice terminal commands on a VM
without risking your main OS



Testing Code Across OSes

Developers check their app works on
Windows, macOS, and Linux VMs



Isolated Experiments

Run suspicious files or try new tools in a
disposable, sandboxed VM

Demo: Setting Up a Virtual Machine

- ▶ **1. Hypervisor Setup:** Install a type-2 hypervisor such as Oracle VirtualBox, VMware Player, or enable Hyper-V on Windows.
- ▶ **2. Obtain OS Image:** Download the installation ISO file for your target guest OS (e.g., Ubuntu Server).
- ▶ **3. Resource Allocation:** Create a new VM profile and permanently lock away host resources (e.g., allocating 4GB RAM, 2 CPU cores, and a 50GB virtual hard disk).
- ▶ **4. OS Installation:** Mount the ISO, boot the virtual machine, and run through the traditional operating system installation wizard.
- ▶ **WSL Comparison:** Unlike WSL, this traditional VM approach requires strict resource partitioning and heavy hardware virtualization overhead.



Try It Yourself — Then Recap

Before installing an experimental app you don't fully trust, what VM action should you take first, and why?

Your answer:

QUICK RECAP

- A VM is a full “computer” built entirely in software by a hypervisor
- Creating one means allocating RAM/CPU, a virtual disk, and an OS image
- Manage VMs with start, stop, snapshot, clone, and delete

Try It Yourself — Then Recap

Before installing an experimental app you don't fully trust, what VM action should you take first, and why?

Your answer: Before installing the experimental app, you should **take a snapshot of the virtual machine (VM)**.

Why?: A snapshot saves the VM's current state. If the app causes problems, crashes the system, or installs malware, you can quickly **restore the VM to the previous working state** without affecting your main computer.

QUICK RECAP

- A VM is a full “computer” built entirely in software by a hypervisor
- Creating one means allocating RAM/CPU, a virtual disk, and an OS image
- Manage VMs with start, stop, snapshot, clone, and delete

What is Cloud Computing?



Renting computing power, storage, and software over the Internet — instead of buying and maintaining your own servers.

Like electricity, not a generator

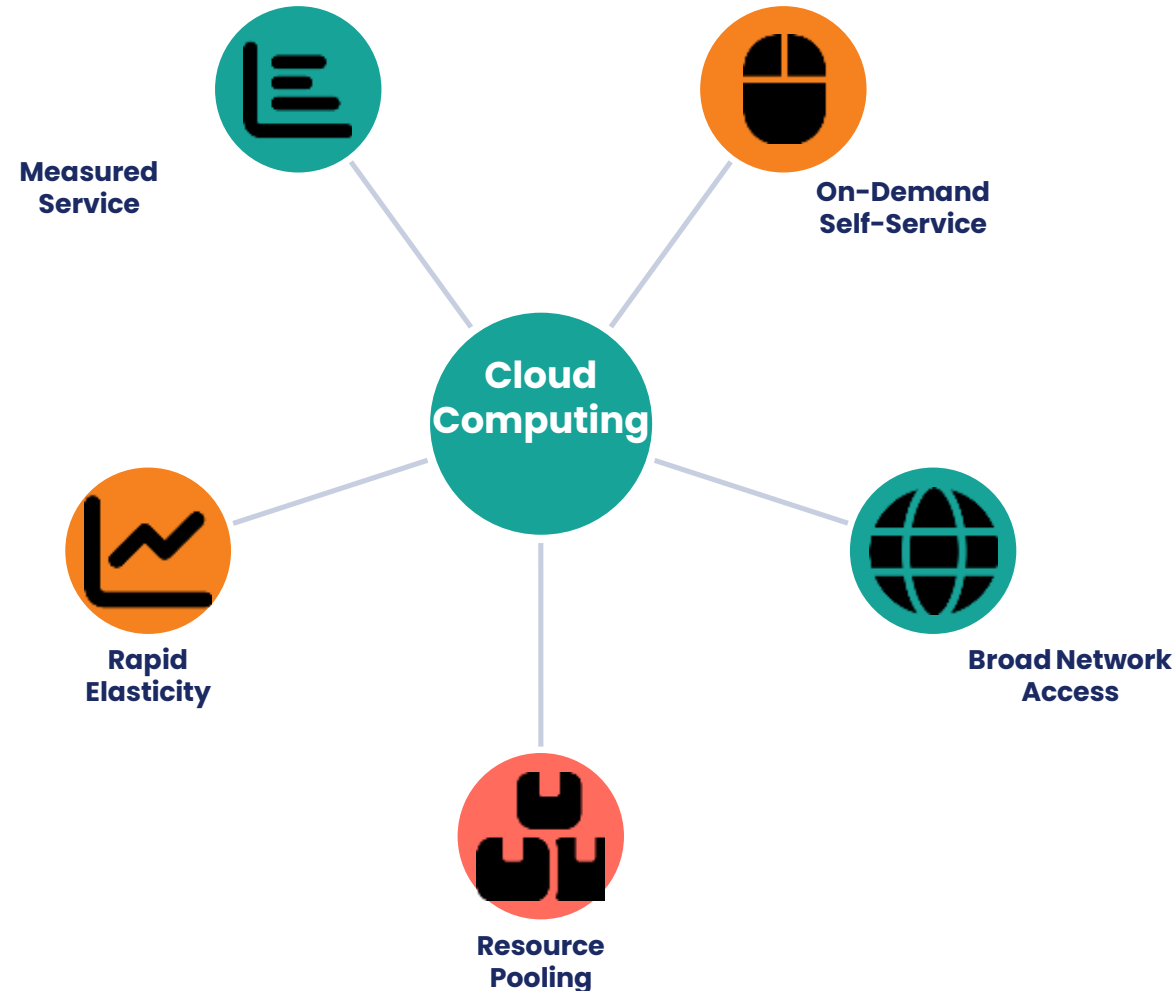
You don't own a power plant to run a fridge — you plug in and pay for what you use. Cloud computing does the same for servers and storage.

Built on virtualization

Cloud providers run millions of hypervisor-managed VMs across giant data centres — everything we just covered, at massive scale.

Key Characteristics of Cloud Computing

The 5 traits (NIST) that make a service genuinely “cloud”



Real World Examples of Cloud Computing



Netflix on AWS

Streams to 300M+ users worldwide
using Amazon Web Services



Google Photos

Stores and syncs billions of photos
across every device you own



Dropbox

Keeps your files backed up and
accessible from anywhere

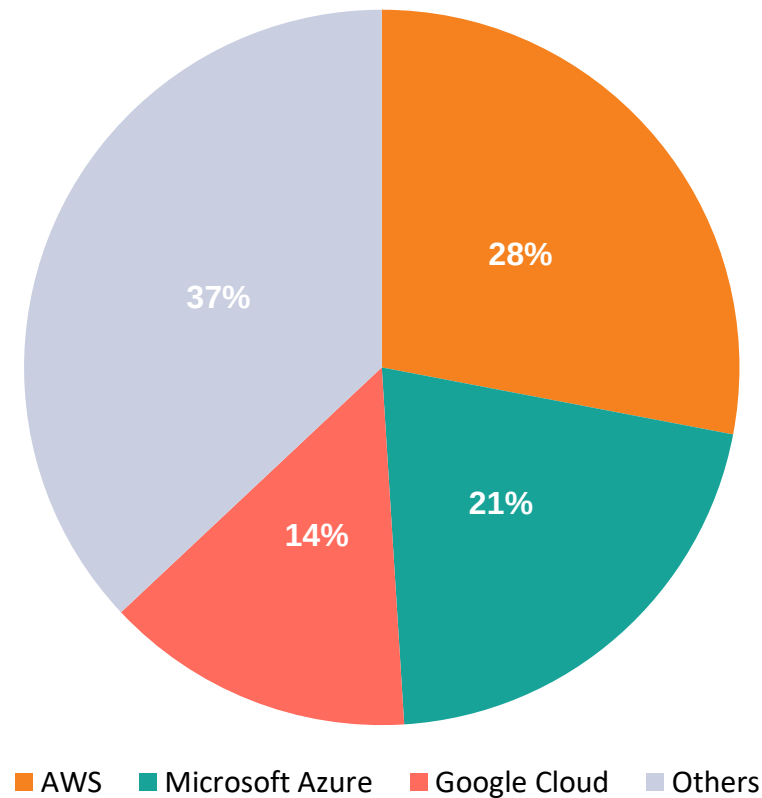


Zoom

Runs video calls for millions of
simultaneous meetings

Who Runs the Cloud? Market Share

Share of global cloud infrastructure spending, Q1 2026



What this means

- Just 3 companies run over 60% of the world's cloud infrastructure
- AWS launched first (2006) and still leads today
- Azure's strength: deep ties to enterprise Microsoft tools
- Google Cloud is the fastest-growing of the “Big Three”

Source: Synergy Research Group, Q1 2026

Try It Yourself: Spot the Characteristic



A shopping site automatically adds more servers during a big sale, then scales back down afterwards, paying only for what it used.

Which NIST cloud characteristic(s) does this describe? Explain briefly:

Your answer:

Try It Yourself: Spot the Characteristic



A shopping site automatically adds more servers during a big sale, then scales back down afterwards, paying only for what it used.

Which NIST cloud characteristic(s) does this describe? Explain briefly:

Your answer:

This describes **two NIST cloud characteristics**:

1. **Rapid Elasticity** – The cloud automatically adds more servers when demand increases and removes them when demand decreases.
2. **Measured Service** – The company pays only for the cloud resources it actually uses.

So, the answer is: **Rapid Elasticity** and **Measured Service**.

QUICK RECAP

Cloud Computing

- Renting computing resources over the internet instead of owning them
- Defined by 5 traits: self-service, network access, pooling, elasticity, metering
- Built on the virtualization & hypervisors we already covered
- AWS, Azure, and Google Cloud dominate the global market

Cloud Deployment Models

Where the cloud infrastructure actually lives



Public Cloud

Shared infrastructure, open to anyone

AWS, Azure, Google Cloud



Private Cloud

Dedicated to one organisation only

A bank's internal cloud



Hybrid Cloud

A mix of public and private, connected

Sensitive data private, apps public



Community Cloud

Shared by organisations with common needs

Government agencies sharing infra

Cloud Service Models

The classic “Pizza as a Service” way to understand IaaS, PaaS, SaaS

On-Premises

Make pizza at home

You buy ingredients, own the oven, do everything

IaaS

Take-and-bake pizza

Provider gives servers & storage; you install the OS & apps

PaaS

Pizza delivery

Provider also manages the OS & runtime; you just add your code

SaaS

Dining at a restaurant

Provider manages everything; you just use the finished app

Who Manages What?

Responsibility shifts from you to the provider as you move right

Layer	On-Prem	IaaS	PaaS	SaaS
Applications	You	You	You	Provider
Data	You	You	You	Provider
Runtime	You	You	Provider	Provider
Middleware	You	You	Provider	Provider
Operating System	You	You	Provider	Provider
Virtualization	You	Provider	Provider	Provider
Servers	You	Provider	Provider	Provider
Storage & Networking	You	Provider	Provider	Provider

Try It Yourself: Which Service Model?



You want to write and deploy code without ever configuring a server, OS, or runtime yourself — the platform just runs it.

Which cloud service model is this, and what's a real example of it?

Your answer:

Try It Yourself: Which Service Model?



You want to write and deploy code without ever configuring a server, OS, or runtime yourself — the platform just runs it.

Which cloud service model is this, and what's a real example of it?

Your answer:

This is **Platform as a Service (PaaS)**.

Why?: With PaaS, the cloud provider manages the **servers, operating system, and runtime**, so you only need to write and deploy your code.

Example: **Google App Engine** (also acceptable: **Heroku** or **Microsoft Azure App Service**).

QUICK RECAP

Deployment & Service Models

- Deployment models: Public, Private, Hybrid, and Community cloud
- Service models: IaaS (infrastructure), PaaS (platform), SaaS (software)
- Like pizza: take-and-bake, delivery, or dining out — less work as you go
- Responsibility shifts from you to the provider as you move up the stack

Getting Started on Google Cloud

- **Step 1: The Portal:** Navigate to cloud.google.com and click on the "Get started for free" button.
- **Step 2: Authenticate:** Sign in securely using your existing Google account credentials to enter the console environment.
- **Step 3: Verification:** Enter your country and provide a valid credit card. *Note: This is strictly to verify identity and block bots. You will not be auto-billed after trial credits expire.*
- **Step 4: Launch Project:** Create a new Cloud Project. You now have access to provision remote Compute Engine VMs, VPC networks, and databases!



PUTTING IT ALL TOGETHER

Computing Environment, in Short

**Operating Systems**

Manages hardware & software; process, memory, file, device, security, UI

**Linux File System**

One tree from /, navigated with pwd, ls, cd, and managed with cp, mv, rm

**Computer Networks**

Connect devices; sized PAN → LAN → MAN → WAN, laid out in topologies

**Network Devices**

Routers connect networks; switches & hubs connect devices within one

**Hypervisor & VMs**

One machine runs many virtual ones — Type 1 bare-metal or Type 2 hosted

**Cloud Computing**

Renting elastic, on-demand computing — IaaS, PaaS, SaaS across public/private cloud

Video References

For setting up Virtual Machines

<https://www.youtube.com/watch?v=nCZcTKFbD2Q>

For setting up GCP account

<https://www.youtube.com/watch?v=NNPtzkCFFBw>

For setting up WSL2

<https://www.youtube.com/watch?v=zZf4YH4WiZo>